Hackathon Problem Statements

Regd No: 25091A3296, 25091A32M3

Problem Statement No: 3

Farmer-to-Market Price Discovery Platform Create a real-time mandi price lookup and comparison system using hash tables and balanced BSTs for fast search across hundreds of markets and commodities. Scrape/aggregate public agri-market datasets and apply trend analysis to alert farmers when prices are favorable for selling. Implement an OOP-based notification and subscription system so farmers can track specific crops. This directly tackles the problem of middlemen exploitation and information asymmetry in rural markets.

Regd No: 25091A32P7, 25091A3273

Problem Statement No: 8

Livestock Disease Outbreak Predictor Build a predictive system for livestock disease outbreaks using time-series analysis of veterinary hospital records, vaccination logs, and seasonal patterns. Apply classification algorithms to flag high-risk zones and livestock categories before an outbreak spreads. Use OOP to model different animal types and their disease susceptibility profiles for extensibility to new diseases. This tool can help veterinary departments proactively allocate vaccines and medical camps in high-risk rural clusters.

Regd No: 25091A32N0, 25091A32M5

Problem Statement No: 38

Waste Collection Route Optimizer Design a garbage truck route optimization system using Travelling Salesman Problem (TSP) heuristics (nearest neighbor, 2-opt) to minimize total distance traveled while covering all designated collection points in a ward. Incorporate bin-fill-level data to prioritize routes dynamically. Use OOP to model trucks, bins, and routes with capacity constraints. This reduces fuel costs and improves waste collection efficiency for municipal corporations struggling with irregular collection schedules.

Regd No: 25091A32E3, 25091A3225

Problem Statement No: 45

Public Grievance Redressal Ticketing System Design a citizen grievance management system using a priority-queue-based ticketing engine that ranks complaints (water, sanitation, roads) by severity and duration pending, ensuring urgent issues are addressed first. Apply basic NLP classification to auto-categorize complaints from free-text descriptions. Use OOP to model tickets, departments, and citizens with status-tracking workflows. This improves transparency and accountability in municipal grievance redressal systems.

Regd No: 25091A3221, 25091A32F2

Problem Statement No: 145

Data Anonymization Tool for Public Datasets Build a data anonymization tool implementing the k-anonymity algorithm to generalize/suppress identifying attributes in datasets before public release, ensuring individual records cannot be re-identified while preserving analytical usefulness. Use OOP to model datasets, quasi-identifiers, and anonymization rules. This supports responsible open-data publishing practices for government and research datasets containing personal information.

Regd No: 25091A32B1, 25091A32B2

Problem Statement No: 148

Digital Footprint Privacy Risk Scorer Create a privacy risk assessment tool using a rule-based weighted analysis engine that evaluates a user's digital footprint (public social media info, data-sharing app permissions) to generate a privacy risk score with actionable recommendations. Use OOP to model risk factors and scoring categories. This promotes digital literacy around personal data privacy, an increasingly critical life skill.